

# Ji Bing Ni

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## EDUCATION

### Purdue University

*Bachelor of Science in Computer Science*

West Lafayette, IN

Aug. 2022 - May 2026

## EXPERIENCE

### Software Developer

Jan. 2026 – Present

*Purdue Vertically Integrated Projects*

West Lafayette, IN

- Enabled secure **Model Context Protocol (MCP)** server access via token-based authentication for HTTP calls
- Extended **geoscience AI assistant** with **account-linked actions** through **web-authenticated MCP sessions**
- Reduced technical debt by refactoring redundant frontend/server XML generation into reusable shared functions
- Improved **MCP server reliability** by expanding **test coverage** between **Fastify server** and **MCP server**

### Software Developer

Jan. 2025 – Aug. 2025

*Purdue University – AIM Research Team*

West Lafayette, IN

- Achieved real-time audio to sheet music alignment with **OSMD API** and **Dynamic Time Warping**
- Developed a music evaluator app with **React Native** and **TypeScript** for improving musicians' performance
- Reduced potential hosting costs by migrating app from **client-server** to a **serverless architecture**
- Enhanced app scalability by **approx. 10%** through creation of **4+ reusable UI components**
- Improved production readiness by **approx. 15%** via **Expo Go** for real-device testing on **iOS** and **Android**

### Systems Programming Teaching Assistant

Jun. 2025 – May 2026

*Purdue University – Department of Computer Science*

West Lafayette, IN

- Led **30+ labs** for **500+ students**, improving debugging efficiency and conceptual understanding in **C/C++** via projects on a **custom shell interpreter**, **malloc**, **Bash scripting**, and **HTTP server development**
- Increased project completion rates by **approx. 20%** for **40+ students** through **20+ office hours**
- Improved comprehension for **40+ students** by **approx. 15%** through grading labs & exams with feedback

### Undergraduate Data Science Researcher

Aug. 2024 – May 2025

*The Data Mine - Purdue University*

West Lafayette, IN

- Reduced call center reliance by building an **AI chatbot** for **Wabash** in an **8-member Agile team**
- Enhanced response quality using **prompt engineering** with an **NLP model** and **machine learning**
- Enabled scalable chatbot interactions by implementing a front-end with **RESTful API** and back-end integration
- Strengthened reliability across **20+ devices** through **5+ API tests** validating connectivity and data accuracy

## PROJECTS

### PokéWorld | *React, Express.js, Node.js, MongoDB, TailwindCSS, Material-UI, REST APIs*

May 2024 – Present

- Developed an interactive Pokémon platform with a full-stack **MERN app** powered by **PokéAPI (1,000+ Pokémon)**, **Pokémon TCG API (150+ sets, 10k+ cards)**, and **TMDB API (20+ movies)** for fan exploration
- Integrated **Google OAuth 2.0** authentication to streamline user sign-in and reduce account creation friction
- Reduced backend maintenance by migrating user data to **MongoDB Atlas** for secure, cloud-hosted persistence
- Built an AI chatbot with **Google Gemini** and **prompt engineering** for real-time site navigation help
- Reduced redundant API calls via **debounced search**, **client-side caching**, and **on-demand pagination**
- Shipped full-stack MERN platform to **Vercel** and **Render** with **MongoDB Atlas** for live production release

### Custom Unix Shell | *C/C++, Lex, Yacc, POSIX*

Jan 2024 – May 2024

- Built a **Unix-like shell in C++** with **command execution**, **pipes**, **I/O redirection**, and **background jobs**
- Enabled advanced features such as **expansions**, **subshells**, and **wildcards** using a **Lex/Yacc-based parser**
- Supported live terminal input with **line editing** and **command history** for an experience similar to **bash**

## TECHNICAL SKILLS

**Languages:** Java, Python, C, C++, JavaScript, TypeScript, SQL, R, HTML, CSS, Tailwind CSS

**Frameworks:** ReactJS, React Native, AngularJS, Express.js, Node.js, MongoDB, PostgreSQL, SQLite, Flask, Bootstrap

**Developer Tools:** Git, GitHub, Docker, Azure DevOps, LaTeX, VS Code, IntelliJ, Eclipse

**Libraries:** pandas, NumPy, Matplotlib, BeautifulSoup, SQLAlchemy